

Name of the Experiment: Verification of Two input Multiplexers (MUX) using NOT, AND, OR gates

Objective: To understand, realize and verify the Data Selectors.

Requirement components and equipment's

1. Trainer board.
2. Connecting wire.
3. Three inputs AND gate (IC-7411)-Two pieces.
4. Two inputs OR gate (IC -7432)- One piece.
5. NOT gate (IC -7404)- One piece.

74LS11

1, 2, 13 = 12

3, 4, 5 = 6

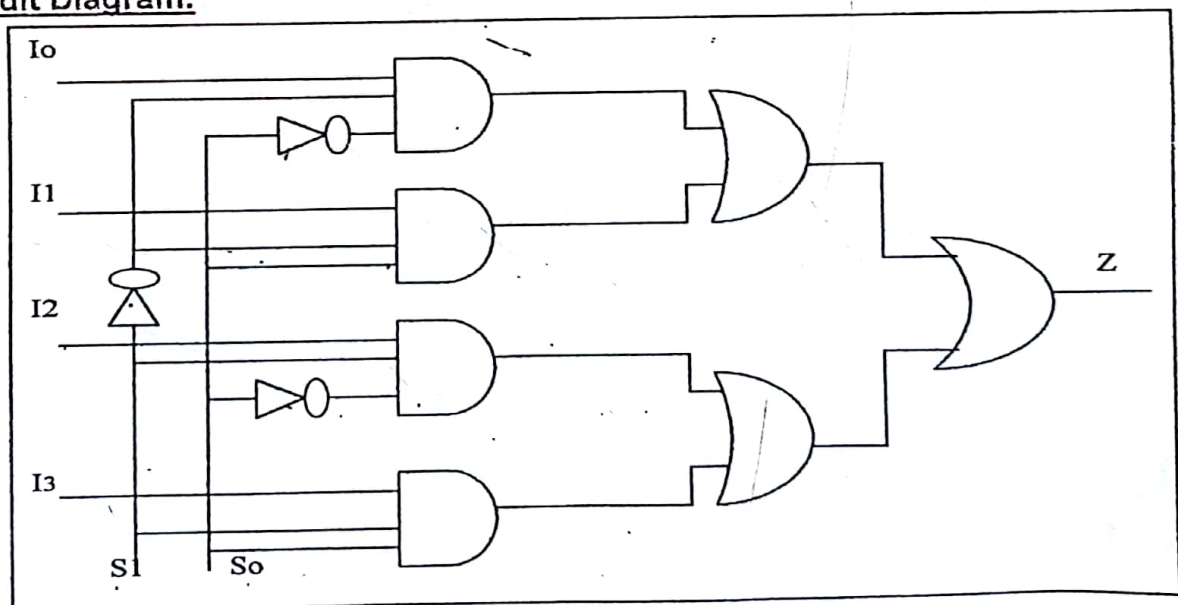
11, 10, 9 = 8

Truth table:

I	S1	So	Output
1	0	0	Z = I0
1	0	1	Z = I1
1	1	0	Z = I2
1	1	1	Z = I3



Circuit Diagram:



Procedure :

- Step 1.** Construct the circuit of fig-1 on the bread board. Every IC's pin "14" connected to +5v position of the DC power supply of Trainer board and pin "7" Connected "GND" position.
- Step 2.** Then we use the pin configuration of the circuit diagram
- Step 3.** For a four input multiplexer we need four data inputs I0, I1, I2, I3 and two select inputs S0, S1.
- Step 4.** The logic level applied to the S input determines which AND gate is enable so that its data input passes through the OR gate to output.

Result :

When S0 and S1 are in the condition of (0,0) then I0 is go to the output.

With S=1, this expression becomes

$$Z = I_0 S + I_1 S + I_2 S + I_3 S \\ = I_0$$

When S=0, this expression becomes

$$Z = I_0 \cdot 0 + I_1 \cdot 1 \\ = I_1$$

Conclusion:

Here we see that the output Z will be high according to the data input signal.